



Your Machine is Infected when...

- Data, such as a spreadsheet or word processor document, gets corrupted, altered or unreadable.
- Programs that you use regularly won't start or operate significantly slower.
- The amount of free hard disk space or free memory is mysteriously disappearing.
- Error messages pop up on your screen.
- The time-date stamp or size of a program file change automatically.
- Your computer becomes extremely slow even while executing normal tasks.
- You find that numerous copies of your files have been created automatically.
- Your computer hangs frequently.
- Your floppies go corrupt quite frequently and you get the message, 'track 0 bad'.
- Your BIOS gives you a message that something is trying to write to your boot sector, and whether it should allow it.
- Any other behaviour that corresponds to your computer not working as it should in normal circumstances.

programmed, it can or cannot damage data, but can always be used to sniff out data such as a credit card number or a password. Trojans come disguised as useful utilities or applications but turn out to be harmful.

The sole aim of a virus or a worm is to cause destruction depending on the 'payload' (the effects of the actions taken by a virus). They are intentionally programmed to infect files, corrupt the installed programs on one's system and some even go to the extent of destroying a hard drive's File Allocation Table. Infected systems are also prone to frequent crashes and slow down significantly. This happens when a virus randomly occupies a memory location that is currently in use by an existing program.

Largely, there are five kinds of viruses: Boot sector, File Infecting, Polymorphic, Stealth and Multi-partite.

Boot sector viruses infect the boot sector (an area that forms the first track of the disk and contains information that is required to boot the computer) of your logical disk (hard disk) or floppy disk, whereas File viruses infect executable files (such as .com, .exe, .ole, etc). One example of a boot sector virus is the Yankee virus, which triggers off at 5 pm, or when [Ctrl] + [Alt] + [Del] is pressed, and plays the tune 'Yankee Doodle Dandy'.

Stealth viruses attempt to hide not only from the operating system but also from the anti-virus software, which makes them extremely difficult to identify. Polymorphic viruses not only hide from the anti-virus software but also keep changing their appearance. In order to detect them, the anti-virus scanner should be able to conduct algorithmic scanning, besides just string-based scanning. Multi-partite viruses are the worst as they can infect the boot sector as well

as executable files. They also possess characteristics of Polymorphic viruses, which makes them extremely lethal. One very popular virus of this kind is Natas. It is a multi-partite polymorphic virus that adds 4,744 bytes to every file it infects.

Just like viruses, worms can be lethal too. A worm is a piece of code that tries to infiltrate another program and then attempts to either destroy the data of the program or keep infecting more files

and programs on the sly. Because of their small size and, nowadays, dependency on JavaScript/Active-X exploits, they can also be embedded in Web pages so that they launch themselves whenever you visit such sites. One example of a worm like this is the popular Nimda ('admin' spelled backwards) worm which spread like wildfire a few months ago and created havoc by crippling a lot of systems worldwide. Nimda took advantage of the loopholes in the Microsoft server, browser and e-mail software to infiltrate PCs.

This new-generation virus can spread in four different ways—via server, via e-mail, through Web pages and by shared folders on the local network.

Trojans

If in the middle of a strenuous task you find that your CD-ROM drive opens and closes by itself at regular intervals, you can bet that you are infected by the Back Orifice Trojan. This is just one of the popular Trojans, and usually most of them go beyond such pranks in order to act as

'Oh no, I've got a Virus! What do I do?'

- Do not panic. It doesn't solve the problem
- Try identifying the virus that you are infected with. Some Trojans and worms can be removed by editing the registry and the Windows startup files. If the virus has come from an e-mail message, delete the e-mail message and the attachments.
- Close all the running programs
- If you are on a network, disconnect yourself from the network. You can log off the network in Windows 98 by pressing [Esc] during the log on process. Pulling off the network cable also helps.
- Try running your anti-virus software. It is likely that even that will be infected.
- In Windows 98 and prior versions you can restart to DOS mode. Try running the command line version of the virus scanner with the clean option.
- Take out the emergency disk/CD that you created from the safe store. Write-protect the emergency disk and boot to DOS using the floppy/CD (make the appropriate changes in the BIOS setup to



enable that). Run the command line version of the virus scanner with the clean option. Use the delete option if the clean option fails.

■ If you are unable to use the virus scanner, but manage to identify the virus, search the Internet for a specific fix for

that virus. Download that fix and run it on your system.

■ If this also fails, take out the hard disk and attach it to your friend's system that has an anti-virus software (hope he's your

best friend). Run the anti-virus software from your friend's system.

■ As a last alternative, copy the data that you have to an alternative partition or another hard disk. Format the hard disk and then load the operating system and the latest version of the anti-virus software with the updates installed. Keep the anti-virus software running in the background and copy the data back selectively. If the virus has infected any of your data files, the anti-virus software should detect it. You will probably only lose data pertaining to that file.